

# P2 - CSG - Sheet and Print

ARC 3133-001 / Architectural Visualization Methods 1 / Fall 2026 - Full Term

**Approach:** Procedural

Revision: 2026-09-23. Instructor review draft. Grading allocation approved; dates remain proposed for review before publication.

## 2.1 - CSG process and instructions

**Introduced:** Week 2 - Sep 1 / **Due:** Week 6 - Sep 29, 11:59 PM Eastern Time / **Weight:** 10%

### Description

Build a CSG object you can still edit, and explain how it is made as a sequence someone else can follow. Use a BIG-style massing diagram: one operation per frame, the same viewpoint throughout and a visible change between steps. Put the diagram sequence and one matching set of instructions (pseudocode) on the same 17 x 11 inch sheet, then make one 3D printed object in P2b.

### Rules & Constraints

- One 17 x 11 inch sheet laid out as a BIG-style process diagram (like Bjarke Ingels Group's step-by-step massing diagrams): six to nine numbered frames that follow your actual sequence, with one clear operation or type of operation per frame.
- Use white building masses, color to identify operations or changes, clear outlines and a simple line-weight hierarchy. Use lighter or dashed lines for hidden edges where needed. Keep the viewpoint consistent; introduce context and vegetation in the final frame. Follow your Graphic Standards Manual.
- Use at least three primitives. Include translation (move), rotation and scale, plus union and difference. Intersection is shown in class and optional here. In each subtraction, label the solid you keep and the cutter.
- Draw the final solid at building scale with a 1.75 m figure, vegetation, ground and shadow.
- Beside the diagrams, add clear instructions and matching pseudocode - one step per operation, with the exposed parameters.
- Refer to Di Mari and Yoo's Operative Design and Di Mari's Conditional Design, and name the spatial operation your sequence explores.
- Submit your editable graph with its exposed inputs and every operation still in place.

### Parameters

- Expose the primitive dimensions, move values, rotation angle/axis/center and scale factor. Name each input by what it does geometrically, and state its units.
- Define the origin and axes, where each primitive starts, and the order of each Boolean. In a difference, identify the solid you keep and the cutter you remove.
- Keep the operations editable. Keep Blender's native node names and use Frames or groups to explain the sequence. Work in Blender Geometry Nodes or an already approved Rhino/Grasshopper route.

### Required Studies

- **Six to nine numbered frames.** Start with at least three primitive solids. Include translation, rotation, scale, union and difference; intersection is optional. Mark what is added and what is removed so each change reads without guessing. Each frame shows one operation or type of operation, with its values.
- **Matching instructions (pseudocode).** Put one instruction beside each frame. Refer to objects by type, position or step number; add names once the sequence is complex enough to need them. State units and axes once, then give enough dimensions, moves and other transforms for someone to rebuild the result. You don't have to name every in-between solid.

- **BIG-style graphic language.** Keep building masses white and use your palette to highlight added, removed or changed geometry. Use strong outer outlines, lighter secondary edges and dashed hidden edges where useful. Watch the [diagram tutorial](#), also listed in Resources. Save vegetation and other context for the final frame.
- **Final solid at building scale.** Show the final geometry with a 1.75 m figure, vegetation, ground and shadow. Keep this presentation context separate from the CSG object you will print. Name the spatial operation you explored, with a reference to Operative Design or Conditional Design.

## Workflow

1. Choose a spatial operation from the readings, sketch a sequence and build it from primitives using an editable graph.
2. Check the order of the Booleans and change an exposed input. Make sure everything after it updates without rebuilding the graph.
3. Write the steps from your working model. Ask a partner to find any size, position or object you left undefined, and fix those instructions.
4. Export the view of each operation from one camera and lay out the single sheet with your Graphic Standards Manual. Make sure every frame matches the current graph.

## Deliverables

- **One 17 x 11 inch landscape PDF sheet:** Lastname\_Firstname\_P2a.pdf. Include all numbered frames, adjacent instructions/pseudocode, exposed values, the final architectural view and source credits.
- **Editable graph:** .blend, or .gh plus .3dm if used. Internalize Grasshopper referenced geometry and include any other files the graph needs. A baked mesh or screenshots alone can't show the operation history.
- Submit in Lastname\_Firstname\_P2a/. Keep the object for the separate P2b print.

## Evaluation

- **Computational understanding - 30%:** explain the transforms, which solids each Boolean uses and in what order; show how an exposed input changes the result.
- **Visual communication - 30%:** the diagram and pseudocode agree, and consistent views explain each change and the final spatial result, following your Graphic Standards Manual.
- **Technical execution - 25%:** the required operations stay editable; the dimensions, axes, operands and instructions are enough to rebuild it; the submission files are complete.
- **Exploration - 15%:** purposeful parameter tests and your chosen sequence support the spatial effect you intended.

## Before You Submit

- [ ] Six to nine frames; at least three primitives; translation, rotation, scale, union and difference. Intersection is optional.
- [ ] Difference frames label the kept solid and cutter.
- [ ] Instructions and pseudocode match the graph and include parameter values.
- [ ] Final view includes a 1.75 m figure, vegetation, ground and shadow.
- [ ] One sheet, and an editable file that opens with everything it needs.

## 2.2 - CSG 3D printed object

**Introduced:** Week 3 - Sep 8 / **Due:** Week 8 - Oct 13, 11:59 PM Eastern Time / **Weight:** 5%

**Preparation checks:** Week 6 - Sep 29: Before you queue the print, check the geometry, units, orientation and sliced preview.

### Description

Print a variant of your P2a CSG object and explain the decisions that make it printable. Compare the physical object with the digital geometry. Record any change you made for closure, feature thickness or orientation, so the print can always be traced back to the original.

### Rules & Constraints

- Print a variant of your CSG object from P2a, within 60 mm on each axis and the 200 g allocation, without supports.
- Hand in the physical print during class on the listed due date; submit digital files and brief fabrication notes by 11:59 PM Eastern Time that night. No additional presentation sheet is required.
- Plan how to avoid supports: change the orientation, split the object into parts to assemble, or adjust overhangs. Bridging is allowed if it prints well on the printer you use.
- Submit the editable source, the exported mesh and the slicer project. Explain any changes you made to the geometry so it would print.

### Parameters

- Uniform print scale and final X/Y/Z dimensions in millimeters. The assembled object must be no larger than 60 mm on any axis; if you print in parts, each part must also fit within 60 mm on each axis.
- Printer and material profile, draft layer height, wall count and infill. Supports must be off. Record the values you chose instead of assuming one standard profile.
- Orientation, the face touching the print bed, estimated time and filament use. The total allocation is 200 g, failed attempts included.

### Required Studies

- **Geometry check.** Show the final mesh as a closed solid with all its normals facing outward. Fix open edges, internal faces and self-intersections before slicing.
- **Orientation comparison.** Compare two positions on the print bed. Label contact areas, overhangs, bridges and floating islands, and explain the orientation you chose. Split the object into parts and assemble them after printing, or adjust overhangs where needed. Bridging is fine if it works with your printer and material; check the sliced preview and test any uncertain spans.
- **Sliced and physical result.** Check the layer preview for thin features that disappear and parts that start in mid-air. Record dimensions, settings, predicted time and material, and a photograph of the finished object. Explain one difference between the prediction and the print.

### Workflow

1. Copy or export the P2a result for printing and keep its source graph intact. Record any changes to the geometry.
2. Export the mesh in millimeters and check all three dimensions in the slicer - STL files don't store units. Scale uniformly to keep the proportions.
3. Do the geometry, orientation and slicing check before you join the print queue. Follow the FabLab's current orientation and machine procedures.
4. Print with supports off, keep a record of the material you use (failed attempts included), and photograph the final object, assembled if printed in parts. Add short notes on how you avoided supports.

## Deliverables

- **One physical CSG object**, assembled if printed in parts, within 60 x 60 x 60 mm, produced without supports and within the 200 g allocation. Bring it to class on the listed due date with your name on a separate label.
- **Short fabrication notes and a photograph** in the submission folder. Record how you avoided supports, why you chose the orientation, any splitting into parts or geometry changes, the dimensions, and the slicer settings, time and material. The 2.1 diagram sheet stays your only CSG presentation sheet; no second sheet is needed.
- **Editable source, exported mesh and slicer project** in Lastname\_Firstname\_P2b/. Include a screenshot showing the dimensions with supports off. A preview or render does not replace the physical object.

## Evaluation

- **Preparation and fabrication strategy - 30%:** the mesh, orientation, part division and sliced layers add up to a print that works within the size, material and no-support limits.
- **Fabrication quality - 30%:** the finished object keeps the intended geometry, with solid features, a controlled finish and accurate assembly if printed in parts.
- **Visual documentation - 25%:** clear photographs and notes follow your Graphic Standards Manual; the source, mesh, slicer project and physical object all match.
- **Testing and revision - 15%:** orientation comparisons, layer checks and any test prints inform your decisions; explain what worked and what you changed.

## Before You Submit

- [ ] Each dimension is at most 60 mm; supports are off; total filament used, failed prints included, is at most 200 g.
- [ ] You have checked that the mesh is closed, its normals face outward, thin features survive and the layer preview is clean.
- [ ] Your fabrication notes include how you avoided supports, why you chose the orientation, the settings and a photograph.
- [ ] Physical object, source, mesh and slicer project are ready for submission.

**Shared standards:** Course software, fabrication, attribution and AI policies apply. Submit through the channel announced in class/Canvas. Test the packaged source from a clean folder. Percentages follow the instructor review draft. This handout uses US Letter; student sheets are 17 x 11 inches.

**Midterm grade:** The midterm grade accumulates the Graphic Standards Manual, CSG sheet and print, module (3.1) and arrays (3.2). These assignments account for 40% of the final course grade. Report the accumulated result as a percentage of those 40 possible course points. The midterm adds no separate assessment weight.